CSA5103 - CRYPTOGRAPHY AND NETWORK SECURITY LIST OF PROGRAMS

EXPERIMENT 5

Aim: To write a Python program to implement the Affine Caesar Cipher using the encryption formula $C = (ap + b)\ 26$.

Algorithm:

Step 1: Start the program.

Step 2: Read the plaintext message from the user.

Step 3: Read the key values a and b, and verify that a is coprime with 26.

Step 4: Convert each plaintext letter into its corresponding numerical value (A = 0, B = 1, ..., Z = 25).

Step 5: Encrypt each letter using the formula C = (a × P + b) mod 26.

Step 6: Convert the encrypted numerical values back into letters and form the ciphertext.

Step 7: Display the ciphertext and end the program.

Program:

```
text = input("Enter the plaintext: ").upper()
a = int(input("Enter the value of a: "))
b = int(input("Enter the value of b: "))
cipher = ""
for ch in text:
    if ch.isalpha():
        p = ord(ch) - ord('A')
        c = (a * p + b) % 26
        cipher += chr(c + ord('A'))
    else:
        cipher += ch
print("Cipher Text:", cipher)
```

Output:

```
Enter the plaintext: HELLO
Enter the value of a: 5
Enter the value of b: 8
Cipher Text: RCLLA
```

Result:

Thus, the Python program to implement the Affine Caesar Cipher using the encryption formula $C = (aP + b)\ 26$was successfully executed, and the plaintext was encrypted into the corresponding ciphertext.